

NET-EXP

Input/output expansion board for NET series control units

User instructions and warnings

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EN

SYMBOLS

The following symbols are used within this manual to indicate potential hazards.

	Important safety notice. Failure to follow these instructions may result in serious injury or damage to property. Failure to do so may result in product malfunction and create a hazardous situation.
	Important safety notice. Contact with live parts can result in death or serious injury.
	Important information for installing, programming or commissioning the product.

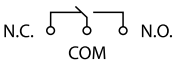
1 PRODUCT DESCRIPTION

The “NET-EXP” and “NET-EXP mini” devices are accessories designed exclusively for pairing with the NET series control units.

NET-EXP allows you to manage up to 8 additional outputs, each of which has a normally closed (N.C.) and normally open (N.O.) output. Additionally, there are 6 additional inputs available. Through the NET series control units, it is possible to customize the operation of each input/output, ensuring maximum adaptability to any type of automation system.

NET-EXP mini is a more compact version that integrates directly onto the NET series control unit (**only NET24N or NET230N**). It offers 1 output and 4 additional inputs. In this case as well, it is possible to customize the operation of each input/output to meet the specific needs of the system.

2 TECHNICAL DATA

For all outputs:		For all inputs:
Contacts configuration		Configurations to choose from: - Free contact; - 8K2 constant resistance.
Rated voltage	NET-EXP 250 V ~ / 30 V  NET-EXP mini 12-24 V ~ / 	
Rated current	Max 2 A (only resistive loads)	

3 ELECTRICAL CONNECTIONS



! Risk of injury and material damages due to electric shock !



! Risk of malfunction due to improper installation !

Make the connections by following the indications of the wiring diagram.

WARNING For adequate electrical safety, all cables must have double insulation. Be sure to keep them clearly separated (**minimum 4 mm in air or 1 mm through additional insulation**) safety extra-low-voltage cables from low-voltage cables (230V ~) by arranging them within plastic raceways and securing them with appropriate clamps near the terminal blocks.

WARNING For the connection to the network, use multipolar cable of minimum section 3x1,5mm² and the type required by current regulations. To connect the motors, use cable minimum section 1.5 mm² and the type required by current regulations. By way of example, if the cable is outside (outdoors), it must be at least equal to H05RN-F while, if inside (in channel), it must be at least equal to H05VV-F.

WARNING Connect to the mains 230 - 240 V ~ 50/60 Hz via an omnipolar switch or other device that ensures omnipolar mains disconnection, with a contact opening distance = 3 mm.

WARNING All cables should be stripped and unsheathed in the immediate vicinity of the terminals. Keep the cables slightly longer so that any excess can be removed later.

WARNING Connect the grounding conductor to the appropriate terminal, taking care to keep its length longer than that of the active conductors so that in the event of the cable coming out of the fixing seat, the active conductors are the first to stretch.

WARNING When connecting the encoder to the control board, use only a dedicated 3x0,22mm² cable.

NET-EXP board terminal table

1	IN 1	EXP_INPUT 1	Configurable Input (See P085 for selectable values)
2	C		
3	IN 2	EXP_INPUT 2	Configurable Input (See P086 for selectable values)
4	C		
5	IN 3	EXP_INPUT 3	Configurable Input (See P087 for selectable values)
6	C		
7	IN 4	EXP_INPUT 4	Configurable Input (See P088 for selectable values)
8	C		
9	IN 5	EXP_INPUT 5	Configurable Input (See P089 for selectable values)
10	C		
11	IN 6	EXP_INPUT 6	Configurable Input (See P090 for selectable values)
12	C		
13	N.O.	EXP_OUTPUT 1 Configurable relay output not powered. Max contact capacity 2A/250V~ /30V === Only resistive loads (See P091 for selectable values)	
14	N.C.		
15	C		
16	N.O.	EXP_OUTPUT 2 Configurable relay output not powered. Max contact capacity 2A/250V~ /30V === Only resistive loads (See P092 for selectable values)	
17	N.C.		
18	C		
19	N.O.	EXP_OUTPUT 3 Configurable relay output not powered. Max contact capacity 2A/250V~ /30V === Only resistive loads (See P093 for selectable values)	
20	N.C.		
21	C		
22	N.O.	EXP_OUTPUT 4 Configurable relay output not powered. Max contact capacity 2A/250V~ /30V === Only resistive loads (See P094 for selectable values)	
23	N.C.		
24	C		
25	N.O.	EXP_OUTPUT 5 Configurable relay output not powered. Max contact capacity 2A/250V~ /30V === Only resistive loads (See P095 for selectable values)	
26	N.C.		
27	C		
28	N.O.	EXP_OUTPUT 6 Configurable relay output not powered. Max contact capacity 2A/250V~ /30V === Only resistive loads (See P096 for selectable values)	
29	N.C.		
30	C		

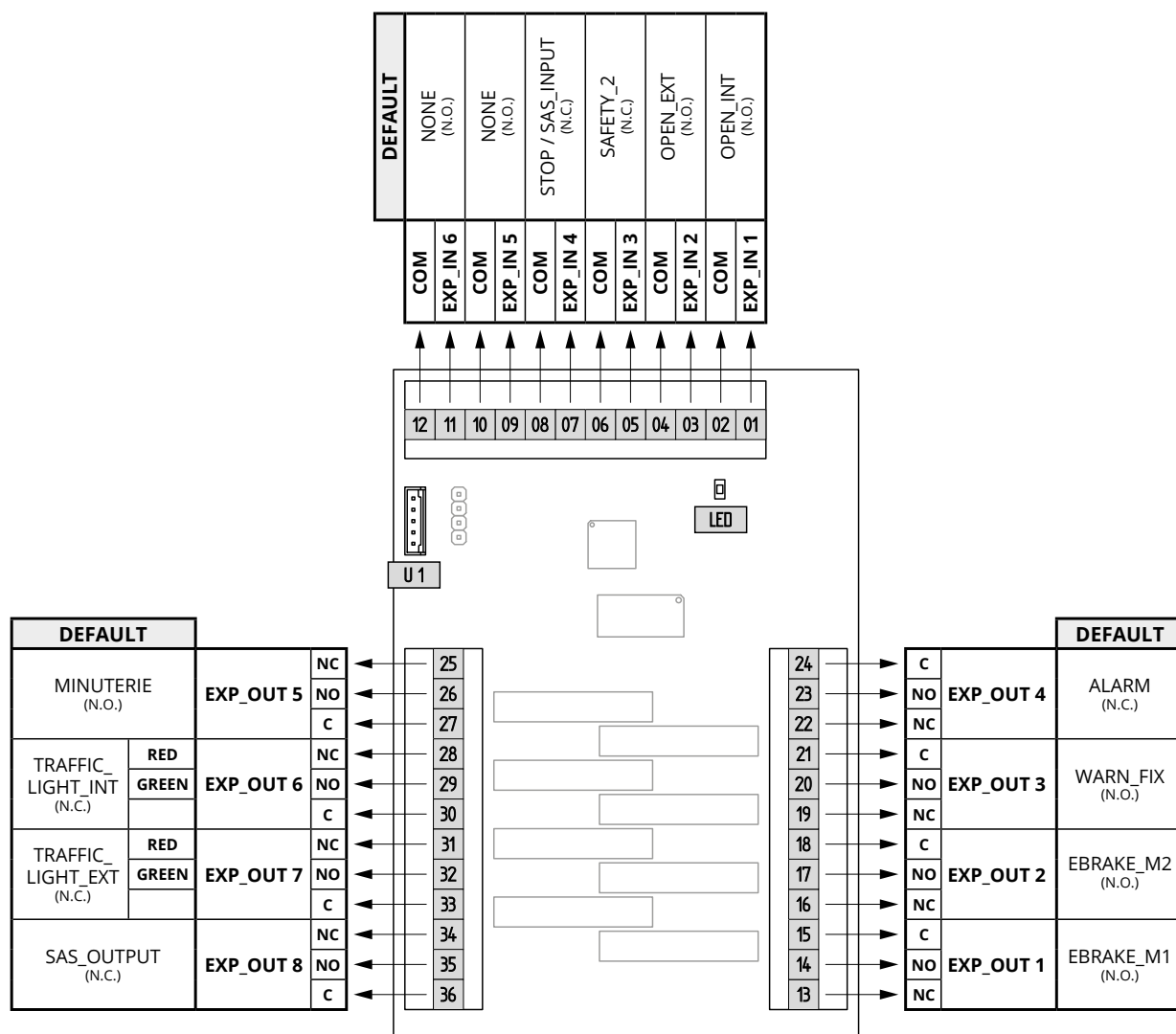
31	N.O.	EXP_OUTPUT 7 Configurable relay output not powered. Max contact capacity 2A/250V~ /30V === Only resistive loads (See P097 for selectable values)
32	N.C.	
33	C	
34	N.O.	EXP_OUTPUT 8 Configurable relay output not powered. Max contact capacity 2A/250V~ /30V === Only resistive loads (See P098 for selectable values)
35	N.C.	
36	C	
U 1		Input module for NET series control units
LED		Status LED

NET-EXP MINI board terminal table

1	N.C.	EXP_OUTPUT 1 Configurable relay output not powered. Max contact capacity 2A/12-24V ~ / === Only resistive loads (See P091 for selectable values)
2	C	
3	N.O.	
4	IN 1	EXP_INPUT 1 Configurable Input (See P085 for selectable values)
5	C	Inputs common
6	IN 2	EXP_INPUT 2 Configurable Input (See P086 for selectable values)
7	IN 3	EXP_INPUT 3 Configurable Input (See P087 for selectable values)
8	C	Inputs common
9	IN 4	EXP_INPUT 4 Configurable Input (See P088 for selectable values)
10	IN 6	EXP_INPUT 6 Configurable Input (See P022 for selectable values)
11	C	Inputs common
12	IN 5	EXP_INPUT 5 Configurable Input (See P021 for selectable values)
13	IN 4	EXP_INPUT 4 Configurable Input (See P020 for selectable values)
14	C	Inputs common
15	IN 3	EXP_INPUT 3 Configurable Input (See P019 for selectable values)
16	IN 2	EXP_INPUT 2 Configurable Input (See P018 for selectable values)
17	C	Inputs common
18	IN 1	EXP_INPUT 1 Configurable Input (See P017 for selectable values)
U 1		Input module for NET series control units
LED		Status LED

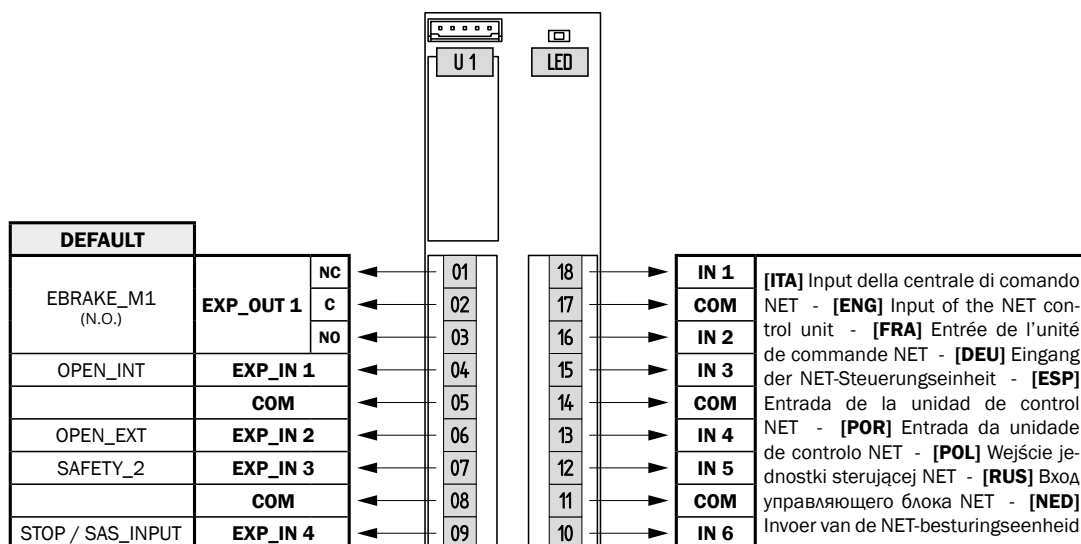
NET-EXP WIRING DIAGRAM

WARNING: All devices connected to the OUTPUT of NET-EXP must necessarily use a suitable auxiliary power source (not provided).

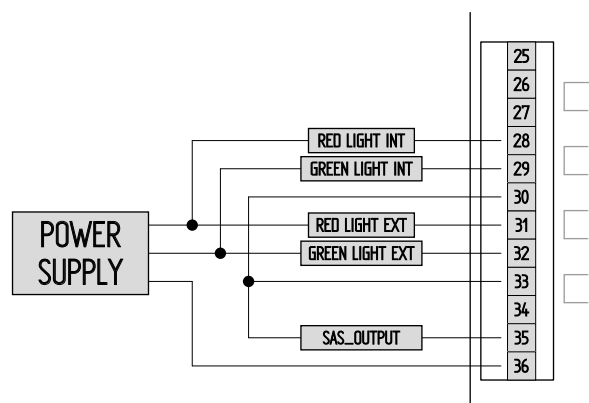


NET-EXP MINI WIRING DIAGRAM

WARNING: All devices connected to the OUTPUT of NET-EXP must necessarily use a suitable auxiliary power source (not provided).



Connection for traffic light off when the gate is closed



WARNING: To ensure proper operation, it is advisable to set the following parameters on the NET control unit: **P053=0** and **P078=1**. It is recommended to refer to the manual of the NET series control unit in use for detailed instructions on how to make these parameter changes.

4 STATUS LEDS DESCRIPTION

Led on: The expansion board power is on and it communicates correctly.

Led off: The expansion board power is off.

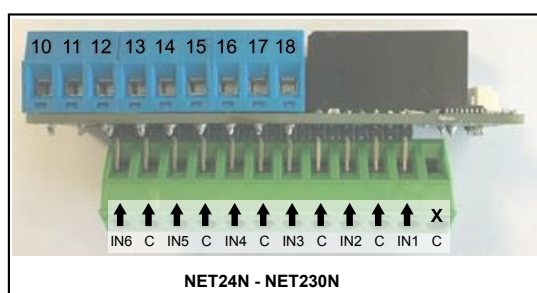
Led flashing: The expansion board power is on but it does not communicate correctly.

5 ENABLING THE NET-EXP BOARD

WARNING The NET-EXP expansion's connection must be made with the NET control panel off.

Follow the instructions below for proper wiring and enabling the expansion board:

1. With the power off, connect (insert in the case of NET-EXP mini, see image on the side) the expansion board to the NET24N or NET230N control unit as indicated and connect the appropriate cable provided. Ensure that the cable is wired correctly.
2. Turn on unit power (the LED turns on flashing).
3. Enable the expansion board by setting parameter P078=1. The LED remains on to confirm correct board communications with the control unit. If flashing, it indicates a communication problem (check correct parameter settings and make sure the connection cable is not damaged or detached).



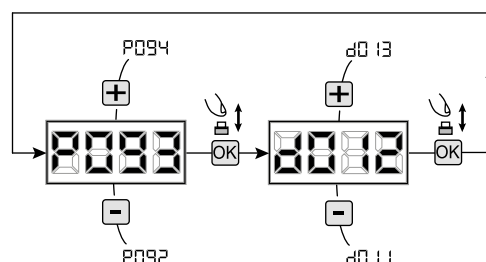
WARNING By enabling the expansion the relative inputs and outputs are already configured for standard installation. Remember to jumper connect any unused NC inputs of the expansion, before any other operation.

WARNING Each time the I/O (P010) settings are reset, the parameters indicated above are changed to default settings (this deactivates the expansion board). Before any other operation, remember to set the parameters again as explained.

6 INPUTS AND OUTPUTS CONFIGURATION

Each input/output may be configured for the desired function if the installation requires different and/or additional controls compared to the standard described in the wiring diagrams.

1. Scroll the parameters with the [+] and [-] buttons until the parameter desired is displayed;
2. Access the parameters (i.e. P093) by pressing the [OK] button;
3. By using the [+] and [-], buttons, set the value corresponding to the desired functions (refer to the "configuration parameters" table on page 22);
4. Confirm the selection by pressing the [OK] button (P093 appears on the display screen).
5. Carry out the connection you have just configured.



WARNING At the end of the programming procedure, use the buttons [+] and [-] until the appearance of the symbol "---", the operator is now ready again for new manoeuvres.

7 DESCRIPTION OF INPUTS / OUTPUTS

The tables shown below describe the operation of all possible selections for both the inputs and outputs present on the board.

INPUTS (IN / EXP_IN)	
Message	Description
NONE	Not used.
START	N.O. input start. In case of action, this results in the opening or closing. It may function in "inversion" (P049=0) or "step-by-step" (P049=1) modality.
PED	N.O. pedestrian start. In case of action, this results in the partial opening of the gate. The adjustment of the pedestrian pace can be set with P043.
OPEN	N.O. input open. In case of action, this results in the partial opening of the gate.
CLOSE	N.O. input close. In case of action, this results in the closing of the gate.
OPEN_PM	Dead man N.O. input open The gate opens for the amount of time in which the button is pressed.
CLOSE_PM	Dead man N.O. input close The gate closes for the amount of time in which the button is pressed.
ELOCK_IN	Electric lock activation N.O. input. In case of action, this leads to the activation of the "LOCK" output of the board, see P062.
PHOTO_1	N.C. input. Photocell 1. To select the operating mode, see P050. If not used, bridge the input.
PHOTO_2	N.C. input. Photocell 2. To select the operating mode, see P051. If not used, bridge the input.
SAFETY_1	N.C. input. Safety edge 1. To select the operating mode, see P067. If not used, bridge the input.
SAFETY_2	N.C. input. Safety edge 1. To select the operating mode, see P068. If not used, bridge the input.
STOP / SAS_INPUT	N.C. contact. (SAS_INPUT): If connected to WARN_FIX / SAS_OUTPUT of a second control unit, causes the "bank door" to operate (disabling the opening of the second door until the first is not fully closed). Stop N.C. input. In case of action, this stops the movement during any manoeuvre. If this is not used, bridge out the connection.
OPEN_INT	Internal OPEN command, causes the internal traffic light to switch when the gate reaches its open position. If an OPEN_EXT command is also received before the gate starts to close it will process this before the automatic close time begins.
OPEN_EXT	External OPEN command, causes the external traffic light to switch when the gate reaches its open position. If an OPEN_INT command is also received before the gate starts to close it will process this before the automatic close time begins.
AUX_IN	Input for the control of the AUX_OUT output.
FCA_1	Motor 1 opening end stop N.C. input If not used, disable the input with the corresponding parameter.
FCC_1	Motor 1 closing end stop N.C. input If not used, disable the input with the corresponding parameter.
FCA_2	Motor 2 opening end stop N.C. input If not used, disable the input with the corresponding parameter.
FCC_2	Motor 2 closing end stop N.C. input If not used, disable the input with the corresponding parameter.
SAFETY_INHIBITION	N.C. input SAFETY inhibition. When open, it causes the by-pass of the SAFETY inputs, which are ignored even if activated.
RESET	N.C. contact. Triggers the resetting of the control unit.

OUTPUTS (OUT / EXP_OUT)	
Message	Description
NONE	Not used.
ELOCK_M1	N.O. contact. Output for M1 motor electric lock.
ELOCK_M2	N.O. contact. Output for M2 motor electric lock.
ELOCK_INV_M1	N.O. contact. Output for M1 motor inverted electric lock (for example for operating the electromagnet of the barriers).
ELOCK_INV_M2	N.O. contact. Output for M2 motor inverted electric lock (for example for operating the electromagnet of the barriers).
WARN_FIX SAS_OUTPUT	N.C. contact. (SAS_OUTPUT): If connected to STOP / SAS_INPUT of a second control unit, causes the “bank door” to operate (disabling the opening of the second door until the first is not fully closed). N.O. contact. (WARN_FIX): Operation as a fixed gate open indicator.
WARN_INT	Flashing gate open light: Flashing output during slow opening and fast closing, always ON when the gate is open, always OFF only at the end of a closing.
FLASH_FIX	N.O. contact. Fixed flashing output.
FLASH_INT	N.O. contact. Intermittent flashing output.
EBRAKE_M1	N.O. contact. Output for motor 1 brake (reversible).
EBRAKE_M2	N.O. contact. Output for motor 2 brake (reversible).
MINUTERIE	N.O. contact. The contact closes for 3 seconds at the start of each manoeuvre.
ALARM	N.C. contact. The contact is always open and closes when the start of a manoeuvre fails because of an activated safety input (PHOTO, SAFETY, STOP). The contact returns to open when a subsequent attempt to start a manoeuvre is successful. In case of a power failure, the contact is closed and, therefore, can be used to generate an alarm.
TRAFFIC_LIGHT_INT	Output from the relay, the N.C contact supplies the red light and the N.O. contact supplies the green light. The light is red when the gate is closed and/or the motors are in motion (if the wiring diagram on page 8 is used, the lights are off when the door is closed). The relay exchanges and the green traffic light are turned on when the gate is open. In case of operation with an internal and an external traffic light, the activation is tripped by the START/OPEN/CLOSE commands and the two traffic lights work in the same way. In case of priority operation (activation with OPEN_INT/OPEN_EXT commands), only the traffic light corresponding to the entrance turns green when the gate is opened, the other traffic light is red. Moreover, after resetting, the traffic lights are red and upon the first command, the unit will search for limit stops/ limit switches during which the traffic lights will remain red until the end of the operation. If clearance time is needed, an automatic closing time (ACT) must be selected with P041. It must be considered that this set time must be at least double that estimated to travel the section between the 2 traffic lights. Therefore, when opened, the traffic lights will remain green for the ACT/2 time. During the second half of the ACT, the traffic lights will remain red to give vehicles enough time to clear the passage. In order to avoid the formation of queues, as long as the traffic light is green, it always accepts the priority command that activated it by starting the ACT over again from 0. Possible booking is effective as soon as the traffic light turns red. The traffic light relative to the other direction will turn green at the end of the ACT.
TRAFFIC_LIGHT_EXT	
AUX_OUT_INPULS	N.O. contact. Output controlled by the AUX_IN input in pulsed mode.
AUX_OUT_STEP	N.O. contact. Output controlled by the AUX_IN input in step-by-step mode.
AUX_OUT_TEMP	N.O. contact. Output controlled by the AUX-IN input in timed mode (the value set with P099 indicates the off delay time in seconds).

8 DETAILED PARAMETER LIST

Configuration Parameters			Default NET24N - NET230N
P078	Enabling the NET-EXP expansion board		000
	Warning: The expansion board is disabled by default. Warning: If a default of settings is carried out, the parameters must be correctly set.		• 000: Disabled • 001: Enabled
P079	EXP_INPUT 1 input type selection		000
P080	EXP_INPUT 2 input type selection		000
P081	EXP_INPUT 3 input type selection		000
P082	EXP_INPUT 4 input type selection		000
P083	EXP_INPUT 5 input type selection		000
P084	EXP_INPUT 6 input type selection		000
	• 000: free contact • 001: constant resistance 8K2		
P085	EXP_INPUT 1		017
P086	EXP_INPUT 2		018
P087	EXP_INPUT 3		016
P088	EXP_INPUT 4		011
P089	EXP_INPUT 5		000
P090	EXP_INPUT 6		000
	• 000: NONE • 001: START • 002: PED • 003: OPEN • 004: CLOSE • 005: OPEN_PM • 006: CLOSE_PM • 007: ELOCK_IN	• 008: PHOTO_1 • 009: PHOTO_2 • 010: SAFETY_1 • 011: STOP • 012: FCA_1 • 013: FCA_2 • 014: FCC_1 • 015: FCC_2	• 016: SAFETY_2 • 017: OPEN_INT • 018: OPEN_EXT • 019: AUX_IN • 020: SAFETY_INHIBITION • 021: RESET
P091	EXP_OUTPUT 1		009
P092	EXP_OUTPUT 2		010
P093	EXP_OUTPUT 3		001
P094	EXP_OUTPUT 4		012
P095	EXP_OUTPUT 5		011
P096	EXP_OUTPUT 6		013
P097	EXP_OUTPUT 7		014
P098	EXP_OUTPUT 8		001
	• 000: NONE • 001: WARN_FIX / SAS_OUTPUT • 002: WARN_INT • 003: FLASH_FIX • 004: FLASH_INT • 005: ELOCK_M1	• 006: ELOCK_M2 • 007: ELOCK_INV_M1 • 008: ELOCK_INV_M2 • 009: EBRAKE_M1 • 010: EBRAKE_M2 • 011: MINUTERIE	• 012: ALARM • 013: TRAFFIC_LIGHT_INT • 014: TRAFFIC_LIGHT_EXT • 015: AUX_OUT_INPULS • 016: AUX_OUT_STEP • 017: AUX_OUT_TEMP
P099	AUX_OUT_TEMP output operation (if activated):		000
	Se=0 disabled; If>0 output controlled by the AUX_IN input in timed mode (the value set indicates the off delay time in seconds).		• 000: "Disabled" • >000: "Timed output (1sec.....255sec)"

9 COMMISSIONING

The testing operation is essential in order to verify the correct installation of the system. **DEA System** wants to summarize the proper testing of all the automation in 4 easy steps:

- Make sure that you comply strictly as described in paragraph "WARNINGS SUMMARY";
- Test the opening and closing making sure that the movement of the leaf match as expected. We suggest in this regard to perform various tests to assess the smoothness of the gate and defects in assembly or adjustment;
- Ensure that all safety devices connected work properly;
- Perform the measurement of impact forces in accordance with the standard EN12453 to find the setting that ensures compliance with the limits set by the standard EN12453.

10 PRODUCT DISPOSAL

DISASSEMBLY

The automation unit must be dismantled by qualified personnel, in accordance with the current accident prevention and safety regulations, and with reference to the installation instructions, but in the reverse order. Before initiating the disassembly operations, disconnect the electrical power and make sure it cannot be reconnected.

DISPOSAL

The automation unit must be disposed of in accordance with the current local and national waste disposal regulations. The product (or its individual parts) must not be disposed of together with other household waste materials.

 **WARNING** In compliance with EU Directive 2012/19/EU on waste electrical and electronic equipment (WEEE), this electrical product should not be treated as municipal mixed waste. Please dispose of the product and bring it to the collection for an appropriate local municipal recycling.

NOTES

[illegible]